

# SANTIAGO ARANGO-PIÑEROS

## *Curriculum Vitae*

(Last updated September 26, 2026)

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| UMass Amherst               | <a href="mailto:santiago.arango.pineros@gmail.com">santiago.arango.pineros@gmail.com</a> |
| Department of Mathematics   | <a href="mailto:sarangopiner@umass.edu">sarangopiner@umass.edu</a>                       |
| LGRT 1238                   | <a href="https://sarangop1728.github.io">https://sarangop1728.github.io</a>              |
| Amherst, Massachusetts, USA | <a href="#">arXiv</a> , <a href="#">MathSciNet</a> , <a href="#">GitHub</a>              |

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### APPOINTMENTS

2025–present     **UMass Amherst**  
Visiting Assistant Professor (Postdoc)

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### EDUCATION

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| 2025 | Ph.D. Mathematics, Emory University<br>Advised by <a href="#">David Zureick-Brown</a> and <a href="#">John Voight</a> . |
| 2019 | M.S. Mathematics, IMPA, Rio de Janeiro, Brazil                                                                          |
| 2017 | B.S. Mathematics, Universidad de los Andes, Bogotá, Colombia                                                            |
| 2017 | B.S. Environmental Engineering, Universidad de los Andes, Bogotá, Colombia                                              |

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### RESEARCH INTERESTS

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| <b>Broad</b>    | Number theory and arithmetic algebraic geometry.                                                                                                                                                                                          |
| <b>Specific</b> | Elliptic curves and abelian varieties, Galois representations, Honda–Tate theory, low degree points on curves, modular curves, generalized Fermat equations, stacky curves, arithmetic statistics, computational and algorithmic aspects. |

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### ARTICLES

11. [Appendix to: On  \$p\$ -adic solubility of  \$Ax^\ell + By^m + Cz^n = 0\$](#) , with Christopher Keyes and Andrew Kobin.

10. *Counting number fields of fixed degree by their smallest defining polynomial*,  
with Fabian Gundlach, Robert J. Lemke Oliver, Kevin J. McGown, Will Sawin, Allechar  
Serrano López, Arul Shankar, and Ila Varma.
9. *Counting primitive integral solutions to spherical generalized Fermat equations*.  
**Canadian Mathematical Bulletin** FirstView (2026).
8. *Fermat descent*.
7. *Galois groups of simple abelian varieties over finite fields and exceptional Tate classes*,  
with Sam Frengley and Sameera Vemulapalli.
6. *Counting 5-isogenies of elliptic curves over the rationals*,  
with Changho Han, Oana Padurariu, and Sun Woo Park. **Journal of the London  
Mathematical Society** 113 (2026), no. 5, e70572.
5. *Bounds for the relative class number problem for function fields*,  
with María Chara, Asimina S. Hamakiotes, Kiran S. Kedlaya, and Gustavo Rama.  
**Journal of Number Theory** 278 (2026), 977–1010.
4. *Galois groups of low dimensional abelian varieties over finite fields*,  
with Sam Frengley and Sameera Vemulapalli. To appear in **Transactions of the  
American Mathematical Society**.
3. *Frobenius distributions of low dimensional abelian varieties over finite fields*,  
with Deewang Bhamidipati and Soumya Sankar. **International Mathematics Re-  
search Notices** 2024 (2024), no. 16, 11989–12020.
2. *Mertens' theorem for Chebotarev sets*,  
with Daniel Keliher and Chris Keyes. **International Journal of Number Theory**  
18 (2022), no. 8, 1823–1842.
1. *The global field Euler function*,  
with Juan Diego Rojas. **Research in the Mathematical Sciences** 7 (2020), no. 3,  
Paper No. 19, 21 pp.

## TEACHING

UMASS AMHERST, INSTRUCTOR OF RECORD

Fall 2026      **Math 714: Arithmetic of Elliptic Curves**

Fall 2026      **Math 132: Calculus II**

Spring 2026      **Math 411: Introduction to Abstract Algebra 1**

Fall 2025      **Math 411: Introduction to Abstract Algebra 1**

ARIZONA WINTER SCHOOL

Spring 2024      Study Group Leader at **AWS 2024**

Fall 2023      Problem Set Leader at **PAWS 2023**

## EMORY UNIVERSITY, INSTRUCTOR OF RECORD

Fall 2023 Math 111: Calculus I

Fall 2022 Math 111: Calculus I

## EMORY UNIVERSITY, TEACHING ASSISTANT

Spring 2022 Math 116: Calculus for life sciences

Fall 2021 Math 221: Linear Algebra

## UNIVERSIDAD DE LOS ANDES, TEACHING ASSISTANT

Spring 2020 Mate 1203: Cálculo Diferencial

Fall 2019 Mate 1203: Cálculo Diferencial

Spring 2019 Mate 1207: Cálculo Vectorial

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INVITED SEMINAR TALKS

- 2025 Harvard University, Number theory seminar
- 2025 Wesleyan University, Algebra seminar
- 2025 Amherst College, Five college number theory seminar
- 2025 University of California San Diego, Number theory seminar
- 2024 University of Illinois Chicago, Number theory seminar
- 2024 Tufts University, Number theory seminar
- 2024 Boston University, Algebra and number theory seminar
- 2024 Brown University, Algebra seminar
- 2024 Emory University, Algebra and number theory seminar
- 2024 Amherst College, Five college number theory seminar
- 2024 Dartmouth College, Algebra and number theory seminar
- 2024 University of Georgia, Athens, Algebra and number theory seminar
- 2023 University of South Carolina, Number theory seminar

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ORGANIZATION

## SPECIAL SESSIONS

- 2026 **Arithmetic Statistics of Rational Points and Number Fields**, AMS Spring Eastern Sectional Meeting, Boston College.  
With Niven Achenjang and Sameera Vemulapalli.

2024 **Explicit Computation with Stacks** (MRC special session), Joint Mathematics Meetings, San Francisco.  
With Jonathan Love and Sameera Vemulapalli.

UMASS AMHERST

2025–present **Five College Number Theory Seminar**, co-organizer

EMORY UNIVERSITY

2024–2025 **Algebra and Number Theory Seminar**, main organizer

2022–2024 Graduate student algebra and number theory seminar, co-organizer

## REFeree WORK

Mathematische Zeitschrift, Proceedings of the American Mathematical Society, Rocky Mountain Journal of Mathematics, Sixteenth Algorithmic Number Theory Symposium.

## SELECTED CONFERENCE AND WORKSHOP PARTICIPATION

2025 Algebraic points on curves, ICERM, Providence, RI.

2024 Nilpotent counting problems in arithmetic statistics, AIM, Pasadena, CA.

2024 Number theory in the Americas 2, Casa Matemática Oaxaca, Oaxaca, México.

2024 XVI Algorithmic Number Theory Symposium. MIT, Boston, MA.

2024 The Mordell conjecture 100 years later. MIT, Boston, MA.

2024 Hypergeometric motives in the LMFDB. MIT, Boston, MA.

2024 Shimura curves in the LMFDB. Dartmouth, Hanover, NH.

2024 Arizona Winter School: Abelian Varieties. Tucson, AZ.

2023 Palmetto Number Theory Series XXXVII. UGA, Athens, GA.

2023 LuCaNT: LMFDB, Computation, and Number Theory. ICERM, Providence, RI.

2023 MRC: Explicit computations with stacks. Buffalo, NY.

2023 Conference in Arithmetic Statistics. CIRM, Marseille, France.

2023 Spring school in Arithmetic Statistics. CIRM, Marseille, France.

2023 Arizona Winter School: Unlikely Intersections. Tucson, AZ.

2023 Introductory Workshop: Diophantine Geometry. MSRI, Berkeley, CA.

2023 Connections Workshop: Diophantine Geometry. MSRI, Berkeley, CA.

2022 Palmetto Number Theory Series XXXV. U of SC, Columbia, SC.

2022 AGNES: Summer school in higher dimensional moduli. Brown, Providence, RI.

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| 2022 | PCMI: Number theory informed by computation. Park City, UT.          |
| 2022 | CTNT: Connecticut summer school in number theory. UCONN, Storrs, CT. |
| 2022 | GAGS: Georgia Algebraic Geometry Symposium. Emory, Atlanta, GA.      |
| 2022 | Arizona Winter School: Automorphic forms beyond $GL_2$ . Tucson, AZ. |
| 2021 | PCMI: Inverse Galois Problem. Online.                                |

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## SOFTWARE AND DATABASES

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| 2023 | <i>L</i> -functions and Modular Forms Database (LMFDB), <a href="https://www.lmfdb.org">https://www.lmfdb.org</a><br>I have made modest contributions. Most recently: <ul style="list-style-type: none"><li>• I developed the Zigzag pictures for the hypergeometric motives pages. See this <a href="#">random family</a>.</li><li>• I updated the Newton polygon pictures for abelian varieties over finite fields, see this <a href="#">random isogeny class</a>.</li></ul> |
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## SKILLS

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| <b>Language</b> | Spanish (native speaker), English, Portuguese. |
| <b>Computer</b> | Python, Magma, SageMath.                       |

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## AWARDS

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| 2025 | Graduate Student Research Award, Emory University Math Department |
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